Appendix B: Hardware Specifications

Appendix B: Standard Hardware Specifications

B.1 Metric Fastener Grades and Properties (ISO 898-1)

B.1.1 Property Classes

Grade	Material	Proof Load (MPa)	Tensile Strength (MPa)	Yield Strength (MPa)	Hardness (HB)	Marking
4.6	Low carbon steel	225	400	240	114-238	4.6 on head
4.8	Low carbon steel	310	420	340	114-238	4.8 on head
5.8	Low/medium carbon steel	380	520	420	147-238	5.8 on head
8.8	Medium carbon steel (quenched)	600	800	640	238-342	8.8 on head
9.8	Medium carbon steel (quenched)	650	900	720	269-342	9.8 on head
10.9	Alloy steel (Q&T)	830	1,040	940	304-361	10.9 on head
12.9	Alloy steel (Q&T, high strength)	970	1,220	1,100	353-415	12.9 on head

Notes: - First number x 100 = minimum tensile strength (MPa) - Second number x 10 x first number = proof load (MPa) - Example: 8.8 -> 8x100 = 800 MPa tensile, 8x8x10 = 640 MPa proof load

Common Applications: - **4.6/4.8:** Non-critical fasteners, light-duty brackets, cable management - **8.8:** Standard structural bolts, machine assembly (most common grade) - **10.9:** Motor mounts, spindle clamps, high-stress joints - **12.9:** Critical fasteners requiring maximum strength, preload applications

B.1.2 Metric Bolt Torque Specifications (Grade 8.8, Lubricated)

Lubrication condition: Light oil or anti-seize compound (friction coefficient mu = 0.12-0.15)

Size	Pitch (mm)	Proof Load (kN)	Torque 75% Proof (N*m)	Torque 85% Proof (N*m)	Clamping Force @ 75% (kN)	Preload Recommended
M3	0.5	1.96	1.0	1.2	1.47	75% for general use
M4	0.7	3.54	2.1	2.5	2.66	75% for general use
M5	0.8	5.52	3.8	4.5	4.14	75% for general use
M6	1.0	8.02	6.4	7.5	6.02	75% for general use
M8	1.25	14.2	15	18	10.7	85% for critical joints
M10	1.5	22.6	30	36	17.0	85% for critical joints
M12	1.75	32.7	52	62	24.5	85% for critical joints
M16	2.0	58.6	130	155	44.0	85% for critical joints
M20	2.5	92.0	260	310	69.0	85% for critical joints

Torque Formula: [T = K d F]

where: - (T) = torque (N*m) - (K) = nut factor (0.18-0.22 for lubricated, 0.25-0.35 for dry) - (d) = nominal diameter (m) - (F) = clamping force (N)

Example Calculation for M8 Bolt:

Proof load = 14.2 kN, Target preload = 75% = 10.65 kN

[T = 0.20 $\times$0.008 $\times$10,650 = 17 N*m]

Tightening Procedure: 1. Clean threads, apply light oil or anti-seize 2. Hand-tighten until snug 3. Use calibrated torque wrench, tighten in star pattern (for multiple bolts) 4. Torque to specified value in 2-3 steps (50% -> 75% -> 100%) 5. Mark bolt head position, verify no rotation after 24 hours (checks for settling)

B.1.3 Metric Nut Types and Grades

Type	Grade	Height	Strength	Applications
Hex Nut (DIN 934)	8, 10	0.8xd	Standard	General purpose, most common
Heavy Hex Nut (DIN 6330)	10	1.0xd	High strength	High-preload joints, vibration
Nylon Insert Lock Nut (DIN 985)	8, 10	0.8xd + nylon ring	Medium	Prevents loosening, single-use recommended
All-Metal Lock Nut (DIN 980)	10	1.0xd	High	High-temperature, reusable lock nut

Type	Grade	Height	Strength	Applications
Thin Jam Nut (DIN 439)	5, 6	0.4xd	Low	Locking second nut, limited space
Square Nut (DIN 557)	5, 6	0.7xd	Low	Captive nuts, T-slots

Lock Nut Comparison:

Type	Reusable	Max Temp (°C)	Vibration Resistance	Cost
Nylon Insert	3-5 times	120	Excellent	$
All-Metal Prevailing Torque	10+ times	300	Good	$$
Split Lock Washer	Multiple	200	Fair	$
Nord-Lock Wedge Washer	100+ times	300	Excellent	$$$

B.2 Imperial Fastener Grades (SAE/ASTM)

B.2.1 SAE Bolt Grades

Grade	Tensile Strength (ksi)	Proof Load (ksi)	Yield Strength (ksi)	Head Markings	Material
Grade 2	60	33	57	No marks	Low/medium carbon steel
Grade 5	120	85	92	3 radial lines	Medium carbon steel, quenched
Grade 8	150	120	130	6 radial lines	Medium carbon alloy, Q&T
A325	120	85	92	A325 marking	Structural bolt (buildings, bridges)
A490	150	120	130	A490 marking	High-strength structural bolt

Note: 1 ksi (kilopound per square inch) = 6.895 MPa

B.2.2 Imperial Bolt Torque Specifications (Grade 5, Lubricated)

Size (UNC)	TPI	Torque 75% Proof (lb*ft)	Torque 85% Proof (lb*ft)	Clamping Force @ 75% (lbf)
1/4”-20	20	8	9	1,990
5/16”-18	18	17	19	3,150
3/8”-16	16	30	35	4,800
1/2”-13	13	75	85	9,350
5/8”-11	11	150	170	14,750
3/4”-10	10	265	300	21,200
1”-8	8	590	665	37,650

B.3 Washer Specifications

B.3.1 Flat Washer Dimensions (Metric, DIN 125 Form A)

Bolt Size	Inner Dia (mm)	Outer Dia (mm)	Thickness (mm)	Purpose
M3	3.2	7	0.5	Distribute load, prevent marring
M4	4.3	9	0.8	Distribute load
M5	5.3	10	1.0	Distribute load
M6	6.4	12	1.6	Distribute load
M8	8.4	16	1.6	Distribute load
M10	10.5	20	2.0	Distribute load
M12	13.0	24	2.5	Distribute load

Fender Washer (Large OD): Used when clamping soft materials (wood, plastic) or oversized holes. OD = 3-4x bolt diameter typical.

B.3.2 Spring Lock Washer (Split Washer)

Function: Prevent loosening via spring tension and sharp edges biting into surfaces.

Limitations: - **Not effective under vibration** (studies show no improvement vs. flat washer) - **Obsolete for critical applications** (replaced by nylon lock nuts, thread lockers) - **Acceptable for non-critical, low-vibration fasteners**

Better alternatives: - Nylon insert lock nuts (vibration) - Thread locker (medium strength Loctite 243) - Safety wire (aircraft/racing applications) - Nord-Lock washers (high-load, vibration-critical)

B.3.3 Belleville Spring Washer (Disc Spring)

Function: Maintain preload under thermal expansion, creep, or settling.

Applications: - Bearing preload adjustment - Vibration isolation mounts - Maintaining bolt tension in joints subject to thermal cycling

Design Parameters: - Load vs. deflection is non-linear (see manufacturer curves) - Can be stacked in series (more deflection) or parallel (more load) - Material: Spring steel (65Mn), stainless 301

B.4 Threaded Inserts and Fasteners for Aluminum/Plastic

B.4.1 Helicoil Thread Inserts

Purpose: Create steel threads in soft materials (aluminum, magnesium, plastic).

Installation: 1. Drill oversize hole (e.g., 6.6mm for M6 Helicoil) 2. Tap with special STI tap (larger than standard M6 tap) 3. Install Helicoil with installation tool (coiled spring insert) 4. Break off tang with punch

Advantages: - 2-3x stronger than tapped aluminum threads - Repairable (extract damaged Helicoil, install new one) - Standard bolts fit (no special hardware)

Length: 1.5x to 2.5x bolt diameter recommended for full strength

B.4.2 Rivnuts (Rivet Nuts, Nutserts)

Purpose: Install threaded fastener in sheet metal or thin-walled material.

Types: - **Closed-end (sealed):** Prevents leakage, used in waterproof applications - **Open-end:** Lower cost, easier installation

Installation: Requires rivnut tool (manual or pneumatic). Bolt pulls mandrel, collapsing body to form bulge on backside.

Load capacity: 1/3 to 1/2 of standard nut (due to thin wall). Use larger size or multiple rivnuts for high loads.

B.4.3 PEM Nuts and Standoffs (Self-Clinching)

Purpose: Permanent threaded fastener in sheet metal (0.5-3mm thickness).

Installation: Press-fit using arbor press or manual tool. Knurls/undercuts bite into parent material and deform to lock in place.

Advantages: - Flush or near-flush on one side - Very strong (resists pullout better than rivnut) - Reusable threads

Common types: - S-type nuts (flush on one side) - SO-type standoffs (spacers between sheets) - FH-type standoffs (flush head)

B.5 Thread Locking and Sealing

B.5.1 Thread Locker (Anaerobic Adhesive)

Type	Strength	Removable	Max Gap Fill	Temp Range	Applications
Loctite 222 (Purple)	Low	Hand tools	0.15mm	-55 to 150°C	Adjustment screws, set screws
Loctite 243 (Blue)	Medium	Hand tools	0.15mm	-55 to 150°C	General fasteners, vibration resistance

Type	Strength	Removable	Max Gap Fill	Temp Range	Applications
Loctite 271 (Red)	High	Heat (250°C)	0.15mm	-55 to 180°C	Permanent assembly, bearing retaining
Loctite 2701 (Green)	High (wicking)	Heat	0.10mm	-55 to 180°C	Pre-assembled fasteners (apply after tightening)

Curing: Anaerobic (cures in absence of air, presence of metal). 24 hours for full cure, handling strength in 10-30 minutes.

Removal: - Low/Medium: Heat to 120°C with heat gun, disassemble - High: Heat to 250°C (torch), disassemble immediately while hot

B.5.2 Thread Sealant (PTFE Tape vs. Liquid)

PTFE Tape (Teflon Tape): - Wrap 3-4 turns clockwise (viewing end of fitting) - Used for NPT pipe threads (water, air, coolant) - Does NOT provide thread locking (only sealing)

Liquid Thread Sealant (Pipe Dope): - Applied to male threads - Seals better than tape for tapered threads - Some formulations include thread locker (check datasheet)

Hydraulic Fittings (O-ring Face Seal, JIC, SAE): Do NOT use thread sealant. Seal occurs at mating face (O-ring or cone), not threads. Thread locker OK if anti-seize not required.

B.6 Retaining Rings (Circlips)

B.6.1 Internal vs. External Retaining Rings

Type	Function	Installation	Groove Depth
Internal (DIN 472)	Retain bearing in housing bore	Expand ring, insert into groove	0.8-1.2mm typical
External (DIN 471)	Retain bearing on shaft	Compress ring, snap into groove	0.8-1.2mm typical

Material: Spring steel (1.4310 stainless common), phosphate or zinc-plated carbon steel

Load Rating: Axial load capacity 50-80% of groove cross-sectional area x material yield strength. Use shoulder for high loads.

B.6.2 Spiral Retaining Rings

Advantages over stamped rings: - No ears (flush with shaft/bore) - 360° contact (no stress concentration) - Reusable many times

Disadvantages: - Lower load capacity than stamped rings - Requires special installation/removal tool

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B.7 Dowel Pins for Precision Alignment

B.7.1 Dowel Pin Types

Type	Tolerance	Hardness	Applications
Standard Tolerance (m6)	-0.012 to -0.030mm	58-65 HRC	General precision alignment
Close Tolerance (h7)	+0.000 to -0.012mm	58-65 HRC	High-precision jigs, fixtures
Oversize (0.1mm, 0.2mm, 0.5mm)	Standard	58-65 HRC	Repair of worn holes

Material: Ground and hardened alloy steel (1.4305 stainless available)

Sizing: - Hole reamed to H7 tolerance (precision reamer required) - Pin pressed in (1-3 ton press for 6mm x 30mm pin typical) - Interference fit: 0.010-0.020mm for steel, 0.005-0.015mm for aluminum

Reaming depth: Dowel pin length + 5mm minimum (prevents bottoming out)

B.7.2 Spring Pins (Roll Pins, Slotted Spring Pins)

Function: Shear pins, quick-disassembly alignment.

Installation: Drive in with hammer or press. Slot compresses, spring force retains pin.

Advantages: - Compensates for hole tolerance (oversized hole OK) - Removable (vs. solid dowel pin)

Disadvantages: - Lower shear strength than solid dowel pin (30-50% typical) - Can walk out under vibration without retaining method

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B.8 Set Screws (Grub Screws)

B.8.1 Set Screw Point Types

Point Type	Best Use	Holding Power	Damage to Shaft
Cup Point	General purpose		Moderate (indents shaft)
Cone Point	Permanent assembly		High (sharp point mars surface)
Flat Point	Soft shafts		Low (broad area)
Dog Point	Machined detent		None (seats in drilled hole)
Oval Point	Delicate shafts		Low (rounded contact)

Recommendation for CNC: - **Shaft collars, pulleys:** Cup point or cone point (most common) - **Fine-positioning:** Flat point with thread locker - **Precision:** Dog point (shaft has mating detent hole)

B.8.2 Torque Specifications for Set Screws

Grade 45H (Alloy steel, hardened to 45 HRC min):

Size	Torque (N*m)	Hex Key Size
M3	1.0	2.0mm
M4	2.2	2.5mm

Size	Torque (N*m)	Hex Key Size
M5	4.4	3.0mm
M6	7.5	4.0mm
M8	18	5.0mm
M10	36	6.0mm

Installation: - Clean threads, apply medium-strength thread locker - Orient setscrews 90° apart (if multiple per collar) - Torque to spec with hex key (not Allen key ball-end, which reduces torque transfer)

B.9 Specialty Fasteners for CNC

B.9.1 T-Slot Nuts and Bolts

Drop-in T-Nuts: - Used with aluminum extrusion (80/20, Misumi) - Insert from end of slot or drop into slot opening - M5, M6, M8 threads typical - Ball spring or sliding style

Flanged T-Nuts (Hammer Head): - Rotate 90° after insertion into slot - Higher load capacity than drop-in - Requires access to end of slot for installation

Loading: Cantilever load on T-slot nut: 100-500 N typical depending on size. Use multiple nuts or longer bolt for high loads.

B.9.2 Captive Panel Screws

Quarter-turn fasteners (Dzus, Camloc): - Quick-access panels (electronics enclosures, inspection doors) - 1/4 turn locks, spring-loaded retention - Low clamping force (use gasket for sealing)

Thumb screws: - Tool-less adjustment - Knurled head or wing head - Lower torque capacity (hand-tight only)

B.9.3 Shoulder Bolts (Shoulder Screws, Stripper Bolts)

Function: Precision shaft/pivot with threaded fastening.

Design: - Shoulder: Ground diameter (h7 tolerance), hardened - Thread: Smaller diameter than shoulder, used for retention only - Head: Socket head cap screw style

Applications: - Rotating/sliding joints (shoulder acts as bearing surface) - Precision spacers - Linkage pivots

Sizing: Shoulder diameter clearance fit with mating part (H7/g6 typical). Shoulder length = stack height + 0.5mm clearance.

End of Hardware Specifications Appendix